

HealthConnect Week 6 Advanced Analytics & Decision Support Report

Data Analytics Track | Integration, Advanced Development & Validation

Prepared By: Bada Oluwagbade Olaoluwa

Project: Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Primary objective: Deepen and validate Week 5 findings, integrate Data Science outputs, and translate the combined evidence into decision-support recommendations for HealthConnect.

❖ Executive Summary

Week 6 builds directly on the Week 5 exploratory analysis rather than repeating it. The core no-show KPI was revalidated, lead-time segmentation was refined to align with Data Science work, and the strongest Week 5 factors were tested in combination. Cross-track integration was then completed using model feature outputs, Logistic Regression coefficients, and a model scoring file.

The strongest Week 6 finding is the interaction between booking lead time and previous no-show history. Appointments booked 41–60 days ahead have a 68.5% observed no-show rate; when that long lead time is combined with 2 or more previous no-shows, the observed rate rises to 83.3% (130 no-shows from 156 eligible appointments). Lead time is also the strongest statistical segment-level association in the dataset.

The supplied Logistic Regression scoring file achieved approximately 63.6% accuracy, 63.0% precision, 65.8% recall, 64.4% F1, and 68.9% ROC-AUC at the supplied 0.50 classification threshold. The probability scores show useful ranking behaviour: the lowest score decile recorded about 16.5% actual no-shows, while the highest decile recorded about 77.3%. Final operational risk thresholds are not recommended yet because clinic capacity and the relative cost of false positives versus false negatives have not been defined.

❖ Week 5 → Week 6 Transition

Week 5 foundation	Week 6 improvement
Initial EDA and descriptive KPI reporting	Advanced segment and interaction analysis
Original lead-time groupings	Refined 0–7 / 8–21 / 22–40 / 41–60 bands aligned with Data Science
Independent Analytics findings	Cross-track model-informed validation and interpretation
Single-variable findings	Lead time × history, lead time × reminder, lead time × distance, type × history
Descriptive dashboard	Decision-support dashboard with model validation page

❖ Refined Methodology

The Week 6 methodology follows six stages:

1. Revalidate the eligible analytical population: Attended + No-Show. Cancelled appointments remain a separate operational outcome.
2. Recalculate the primary KPI and confirm the Week 5 baseline.
3. Refine segment definitions to align Analytics and Data Science workstreams.
4. Validate important relationships using sample size, rate differences, chi-square tests and Cramer's V effect sizes.
5. Conduct interaction analysis to identify higher-impact operational segments.
6. Integrate model outputs for business interpretation while keeping model evidence separate from causal claims.

❖ KPI Validation

KPI	Validated result	Interpretation
Total appointments	5,000	Full appointment dataset
Eligible appointments	4,737	Attended + No-Show only
No-shows	2,423	Primary missed-appointment outcome
Attended appointments	2,314	Eligible attended outcome
Overall no-show rate	51.15%	Week 5 baseline confirmed
Cancelled appointments	263	Tracked separately
Cancellation rate	5.26%	Operational outcome, not part of no-show denominator

❖ Advanced Analysis & Validated Findings

Finding	Evidence	Business meaning
Booking lead time	29.5% at 0–7 days → 68.5% at 41–60 days	Strongest ordered pattern; long-lead appointments are the clearest reconfirmation target.
Any previous no-show	57.8% vs 46.3% with no previous no-show	Historical behaviour is useful for prioritising supportive outreach.
High previous no-show history	63.5% for 2+ prior no-shows	Risk becomes more pronounced when repeated history is used.
Long lead + high history	83.3% for 41–60 days + 2+ prior no-shows (n=156)	Highest-priority validated interaction segment.
Distance	60.2% at 20+ km vs 48.6% at <5 km	Travel burden appears relevant, particularly when lead time is long.
Reminder status	54.6% without reminder vs 49.9% with reminder	Promising operational signal, but causal impact is unproven.
Appointment type	Follow-up 54.2% vs General Consultation 49.1%	Follow-up workflow merits review, especially for high-history patients.
Age / day / time	Relatively small differences	Do not use as primary intervention criteria on Week 6 evidence alone.

❖ Statistical Validation

Variable	p-value	Cramer's V	Assessment
Lead-time band	<0.001	0.276	Strongest validated association
Any previous no-show	<0.001	0.113	Meaningful secondary association
2+ previous no-shows	<0.001	0.085	Useful history segment

Variable	p-value	Cramer's V	Assessment
Distance band	<0.001	0.062	Small but meaningful
Reminder sent	0.0038	0.042	Small association
Appointment type	0.038	0.042	Small association
Appointment time	0.677	0.013	Weak / not statistically meaningful
Appointment day	0.382	0.037	Weak
Age group	0.236	0.038	Weak

These tests validate patterns within the supplied fictional dataset. They do not establish causal relationships.

❖ Cross-Track Integration with Data Science

Two Data Science collaborators provided different outputs. One shared a Random Forest candidate summary (feature importance, threshold sweep and partial error analysis). Another shared a refined Logistic Regression coefficient table and a scoring file. Because these may use different preprocessing or test splits, their performance metrics should not be combined or directly ranked unless the underlying evaluation design is confirmed.

Analytics finding	Data Science evidence	Integrated interpretation
Lead time rises sharply with no-show rate	Lead-time variables/features are among the strongest model drivers	Highest-confidence shared operational theme.
Previous no-show history is higher risk	Previous no-show variables have positive predictive influence	History should inform supportive prioritisation.
20+ km group has higher observed rate	Distance contributes to model prediction	Access burden is a secondary risk signal.
No reminder has higher observed rate	No-reminder/channel terms appear in model outputs	Useful intervention hypothesis; do not claim causality.
Follow-up appointments have moderately higher rate	Follow-up contributes modestly in model outputs	Review workflow, especially alongside history.

❖ Logistic Regression Scoring-File Validation

Metric	Result
Accuracy	63.6%
Precision	63.0%
Recall	65.8%
F1 score	64.4%
ROC-AUC	68.9%
True negatives	296
False positives	187
False negatives	165
True positives	318

Probability-score validation shows a meaningful ranking gradient: the lowest predicted-probability decile records about 16.5% actual no-shows, while the highest decile records about 77.3%. This supports the score as a prioritisation signal, but does not by itself determine an operational intervention threshold.

❖ Important Integration Issue

The supplied scoring file uses integer values 0–965 in its `appointment\_id` field, while the original HealthConnect dataset uses identifiers such as HC-00001. Therefore the scoring file cannot be safely joined to the original appointment table by appointment ID. The integer field should be treated as a model/test-row identifier only.

Week 7 requirement: Data Science should re-export the scoring file while preserving the true HealthConnect appointment_id and preferably patient_id. This will allow false-positive and false-negative analysis by reminder status, distance, appointment type, lead time and patient history.

❖ Threshold Decision Support

The model threshold changes both patient coverage and error trade-offs. Lower thresholds increase recall but create more false positives and outreach workload; higher thresholds improve precision but miss more true no-shows. A final threshold should only be selected after stakeholders define the relative cost of a missed no-show, an unnecessary outreach contact and the clinic's practical outreach capacity.

❖ Dashboard Improvement Plan

The Week 5 dashboard should be retained as the overview foundation, but Week 6 should add a second page focused on advanced segmentation and model-informed decision support. Low-value visuals such as Age Group and Day of Week should be de-emphasised from the main decision area because their validated associations are weak.

Dashboard change	Reason
Replace old lead-time bands with 0–7 / 8–21 / 22–40 / 41–60	Aligns Analytics and Data Science and reveals a clearer 68.5% long-lead rate.
Add Lead Time × Previous History matrix	Shows compounding risk and the 83.3% priority segment.
Add Lead Time × Reminder chart	Shows reminder-associated difference within comparable lead bands.
Add Distance visual	Distance is more useful than age/day/time for decision support.
Add a Model Validation page	Makes cross-track integration visible and auditable.
Add model metric cards and probability-decile validation	Translates predictive output into stakeholder-friendly evidence.
Add threshold trade-off visual	Shows operational consequences without prematurely declaring a risk threshold.

12. Evidence-Based Recommendations

- Prioritise reconfirmation for appointments scheduled 41–60 days in advance.
- Test a focused intervention for patients with both long lead times and repeated previous no-shows.
- Use previous no-show history to prioritise supportive outreach, not to deny or restrict access.
- Test reminder timing/channel through a controlled pilot before claiming a causal effect.
- Consider additional support for patients travelling 20+ km, particularly when appointments are booked far in advance.
- Keep age, appointment day and appointment time as contextual variables rather than primary intervention rules.

- Use predictive probability as a ranking signal only until operational thresholds are agreed.

❖ Limitations, Risks & Dependencies

- Fictional and anonymised data limits generalisation to a real clinic.
- Observational analysis does not establish causation.
- Reminder assignment may be non-random.
- Distance contains limited missing values.
- Some shared engineered-feature definitions were described as 'likely' rather than confirmed from source code.
- The Logistic Regression scoring export does not retain the true HealthConnect appointment identifier.
- Different Data Science model outputs may use different train/test configurations.
- Final Low/Medium/High operational risk tiers are not yet defined.

❖ Week 7 Analytical Testing Requirements

7. Obtain a scoring export with true appointment_id and patient_id.
8. Confirm exact engineered-feature definitions from source code.
9. Analyse false positives and false negatives by lead time, reminder status, distance, appointment type and history.
10. Agree an operational cost/capacity framework for threshold selection.
11. Validate Power BI measures against notebook outputs.
12. Test the stability of the long-lead + repeated-history segment.
13. Compare predicted probability ranking with actual no-show outcomes on a consistent holdout population.

❖ Conclusion

Week 6 advances the HealthConnect Analytics work from descriptive reporting to integrated decision support. Lead time and previous no-show history remain the clearest shared signals across Analytics and Data Science. The strongest operational segment identified is the combination of 41–60 day booking lead time and 2+ previous no-shows, with an observed no-show rate of 83.3%. The model scoring output also demonstrates useful risk ranking, but threshold selection and row-level error analysis should remain Week 7 tasks after the data interface is corrected.